Introduction to Derivatives

Assignment 3

1) A trader buys a European call option and sells a European put option. The options have the same underlying asset, strike price, and maturity. Describe the trader's position. Under what circumstances does the price of the call equal the price of the put? 2.5 points

2) In the 1980s, Bankers Trust developed index currency option notes (ICONs). These are bonds in which the amount received by the holder at maturity varies with a foreign exchange rate. The ICON specified that if the yen–US dollar exchange rate, S_T, is greater than 169 yen per dollar at maturity (in 1995), the holder of the bond receives $1,000. If it is less than 169 yen per dollar, the amount received by the holder of the bond is:

$$1{,}000 - \max\left[0,\ 1{,}000\left(\frac{169}{S_T} - 1\right)\right]$$

When the exchange rate is below 84.5, nothing is received by the holder at maturity. Show that this ICON is a combination of a regular bond and two options. 2.5 points

3) An investor owns NVIDIA shares. The share price is $110 and a December put option with a strike price of $90 costs $15. The investor is comparing two alternatives to limit downside risk. The first involves buying December put option contracts with a strike price of $90. The second involves instructing a broker to sell the shares as soon as the share price reaches $90. Discuss the advantages and disadvantages of the two strategies. 2.5 points

4) Consider a 9-month dollar-denominated American put option on British pounds. You are given that: (i) The current exchange rate is 1.43 US dollars per pound. (ii) The strike price of the put is 1.56 US dollars per pound. (iii) The volatility of the exchange rate is $\sigma = 0.3$ per year. (iv) The US dollar continuously compounded risk-free interest rate is 8% per annum. (v) The British pound continuously compounded risk-free interest rate is 9% per annum.

Using a three-period binomial model, calculate the price of the put. 2.5 points